

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Field-Portable, Fast GC/TOFMS

Authors: Jack Syage, Brian Nies, Rick Harkewicz

Institute: Syagen Technology, Inc.

ABSTRACT

This talk reports on a fast gas chromatograph, time-of-flight mass spectrometer (GC/TOFMS) for man-portable field use. The complete system is designed to meet performance, size, weight, power, cost, and ruggedness requirements for a laboratory in the field. The core technology is also adaptable to specific applications including real-time point detection for hazardous chemical releases (e.g., chemical weapons), for biological agent signature identification, and for mobile monitoring platforms (e.g., air, ship, truck). In this talk we present recent progress in integrating a low-power, high-speed GC and show the capability for accurately recording fast GC transients for targeted compound detection using a photoionization (PI) source coupled to a quadrupole ion trap, time-of-flight (QitTof) instrument. The system specifications are 40 lb weight, 2 cu ft. size, and 150 W of power consumption. The instrument records mass spectra at nominally 60 Hz (USA) making it suitable for fast GC. The GC is temperature programmable and can cycle typical samples in 120 to 240 s. Preliminary GC measurements gave baseline resolution with peak widths of about 1 s and detection limits of about 20-100 ppb and 10-50 pg. We also present results on a pyrolysis/GC front end showing distinct signature peaks for biological bacteria and spore detection. This work is being supported by the U.S. Army for chemical/biological defense.